DATASCI 101 - Introduction to AI Applications

Danilo Freire

Fall 2026

Course description

Welcome to DATASCI 101! This is a non-technical introduction to artificial intelligence and how it is changing institutions, work and daily life. We will examine how modern AI systems are built and, just as much, how to question what they produce: where the data comes from, when the systems fail, and whom those failures affect. The classes combine short demonstrations (no programming required) with case studies and work on your own project. Anyone can take the course, whatever your major. The aim is a clear-eyed sense of what AI can do, and what it cannot.

Learning objectives

By the end of term, you should be able to:

- Explain in plain language the main ideas behind today's AI systems.
- Identify the common ways these systems fail, and the data problems that cause those failures.
- Read a news article, product page or policy document about AI and assess whether its claims are credible.
- Design a small, realistic plan for an AI application: the data it requires, how you would evaluate it, and how you would limit its potential for harm.
- Consider the ethical, legal and social questions that arise when AI is deployed.

Course logistics

- **Lectures:** Tuesdays and Thursdays, 2:30-3:20pm.
- **Location:** White Hall, Room 206.
- **Instructor:** Danilo Freire (`danilo.freire@emory.edu`). Office hours by appointment. Please email me to arrange a time.
- **Materials:** Lecture notes, tutorials, and assignment templates are on the course GitHub repository. Students must check the repository often for updates.

Prerequisites and software

There are no prerequisites. Familiarity with spreadsheets, basic statistics or introductory programming is useful but not required. We may discuss some simple Python code snippets in class, but you will not be asked to write any code.

Readings and resources

There is no single textbook. Instead, we read short papers, essays and tutorials, all linked on the course website. The syllabus may change during the term, so please check it periodically. If you would like to explore a topic in more depth, here are some books and resources I recommend:

Books

- Artificial intelligence: a guide for thinking humans by Melanie Mitchell (2020). A clear, non-technical overview of AI concepts and history.
- The master algorithm by Pedro Domingos (2015). A readable introduction to machine learning concepts and applications.
- Co-intelligence: living and working with AI by Ethan Mollick (2024). An accessible guide to working alongside generative AI, with practical advice on its strengths and limits.

- The coming wave: technology, power, and the twenty-first century's greatest dilemma by Mustafa Suleyman and Michael Bhaskar (2023). A widely read argument that AI and synthetic biology will be very hard to contain, written by the co-founder of DeepMind.
- The alignment problem: machine learning and human values by Brian Christian (2020). An exploration of what can go wrong when AI systems are deployed in the real world.
- AI & I: an intellectual history of artificial intelligence by Eugene Charniak (2024). A new book by a leading AI researcher, covering the history of the field from the 1950s to the present day.
- Weapons of math destruction: how big data increases inequality and threatens democracy by Cathy O'Neil (2016). A book on the dangers of unregulated mathematical models.
- Atlas of AI: power, politics, and the planetary costs of artificial intelligence by Kate Crawford (2021). A critical look at the social and environmental impacts of AI technologies.
- Hands-on large language models: language understanding and generation by Jay Alammar and Maarten Grootendorst (2024). A practical guide to large language models.

Online courses

- AI for everyone by Andrew Ng (Coursera). A non-technical introduction to AI concepts and applications. Free to audit, with a paid certificate option.
- Elements of AI by the University of Helsinki. A free, friendly introduction to AI and some of its methods.
- AI essentials by Google (Coursera). This programme is designed for people who want to gain practical AI skills for the workplace with no experience required.
- Google prompting essentials by Google (Coursera). A short course on how to effectively use and design prompts for large language models.
- Radical ideas in AI ethics by Pragmatic AI Labs (edX). A course that discusses AI through the lens of human rights and digital autonomy.

Other resources

- The AI doc: or how I became an apocaloptimist by Daniel Roher and Charlie Tyrell (2026). A

light-hearted documentary featuring many of the field's most prominent voices, with views ranging from very optimistic to very pessimistic about AI. Worth watching.

- In machines we trust. A podcast series by MIT Technology Review about the promises and perils of AI. Very accessible and engaging.
- AI for the rest of us. 25-30 minute episodes focused on explaining AI concepts to non-technical listeners.
- The gradient. A publication that features accessible articles on AI research.
- AI Now Institute. Research institute focused on the social implications of AI.
- AI for Good Lab. A Microsoft research group that highlights how AI can change society for the better.
- The AI incident database. A collection of real-world cases where AI systems have caused harm or failed.
- Teachable machine. A web-based tool by Google that allows users to create simple machine learning models without coding.
- The algorithm. A newsletter by MIT Technology Review that covers the latest developments in AI.

Assessments

- **Problem sets (10) — 50%**. Short conceptual and practical tasks. Submit Jupyter notebooks (`.ipynb`), Word documents, or PDFs. Late submissions incur a 10% penalty per day unless authorised in advance. Collaboration for discussion is allowed but answers must be written independently. List your collaborators when you submit.

- **In-class quizzes (5) — 30%**. Each quiz occupies a full lecture (50 minutes). Questions are handed out on paper in class. You type your answers in a Word document or PDF and upload the file to Canvas by the end of the session. Quizzes are open notes and open web, and you may consult AI tools if you disclose which ones you used. They are individual assessments: discussion with other students is not permitted. Questions refer to specific material covered in class, and your reasoning is what is graded.

- **Final group project — 20%**. Groups of 4–5 choose one of seven domains (healthcare, education, finance, law, creative industries, scientific research, or customer service), test existing AI tools in that domain, and design a new AI application. Deliverables: an optional one-page proposal (not graded), a final report of 8–12 pages, and a one-page infographic. No programming is required. Detailed instructions (PDF).

Grading scale

Grade	A	A-	B+	B	B-	C	D	F
Range	91–100	86–90	81–85	76–80	71–75	66–70	60–65	<60

Final grades are rounded to the nearest point.

Academic integrity and AI use

You are welcome to use generative AI tools such as ChatGPT, Claude or GitHub Copilot to brainstorm or draft, but any such use must be declared on your submissions. You are responsible for the correctness of your answers, and any text or code produced directly by a model must be attributed. The Emory Honour Code applies throughout.

Emory Honour Code

The Emory Undergraduate Academic Honour Code is in effect throughout the semester. The Honour Code applies to any action or inaction that fails to meet the communal expectations of academic integrity. Students should strive to excel in their academic pursuits in a just way with honesty and fairness in mind and avoid all instances of cheating, lying, plagiarizing, or engaging in other acts that violate the Honor Code. Such violations undermine both the individual pursuit of knowledge and the collective trust of the Emory community. Students who violate the Honour Code may be subject to failure of the course, a reportable record, suspension, permanent expulsion, or a combination of these and

other sanctions. The Honor Code may be reviewed at: http://catalog.college.emory.edu/academic/policies-regulations/honor-code.html.

Accessibility

If you require accommodations, please contact the Department of Accessibility Services early in the term and provide their documentation. I will follow DAS guidance for all assessments, quizzes included. Please contact me privately to discuss any specific needs.

Course schedule

The schedule is below. Each class comes with a short description and two reading lists.

- *Suggested readings* are the two to four short, non-technical pieces I recommend for that class. Start with these.
- *Further readings* go deeper. Some are academic papers, some are long reports, and a few are technical. Use them for the topics that interest you most, and for your final project.

The slides are self-contained, and everything the assignments and quizzes ask of you is covered in class. Come talk to me if you would like more on any topic. The schedule may change during the term. Please contact me if anything is unclear.

Module 0 — Orientation

August 27, 2026:

- Syllabus and course repository: https://github.com/danilofreire/datasci101.
- Lecture 01: Course introduction.

Suggested readings:

- Our World in Data (2025). Artificial intelligence. A concise overview of AI trends and applications, with links to further resources.

- Stanford University's Human-Centered Artificial Intelligence (2025). AI index annual report. An excellent report on the state of AI, with many useful charts and references.
- The New Yorker (2026). What is Claude? Anthropic doesn't know, either. A fascinating article about the challenges of building and understanding large language models. Also available here.

Further readings:

- Menlo Ventures (2025). 2025: the state of consumer AI. How AI is being used in consumer products, with examples and market data.
- Morgan Stanley (2025). AI's next leap: 5 trends shaping innovation and ROI. A short text on the current state of AI applications in industry.
- McKinsey & Company (2025). The state of AI: how organizations are rewiring to capture value. A business-focused overview of AI adoption and impact.

September 1, 2026:

- Lecture 02: A brief history of AI and the recent shift.

Suggested readings:

- Dick, S. (2019). Artificial intelligence. *Harvard Data Science Review*, *1*(1). A brief history of AI research and applications. Written for a general audience.
- Pradhan, M. (2023). A non-technical introduction to Transformers. A clear explanation of the transformer architecture behind many modern AI systems.
- Georgia Tech's Polo Club of Data Science (2025). Transformer explainer. An interactive visualisation of how transformers work.
- Whang, O. (2026). We don't really know how AI works. That's a problem. *The New York Times Magazine*. A feature-length, non-technical primer on interpretability research, from sparse autoencoders to Goodfire's work with Prima Mente on Alzheimer's diagnosis. A very accessible introduction to why the "black box" problem matters.

Further readings:

- Grzybowski, A., Pawlikowska–Łagód, K., & Lambert, W. C. (2024). A history of artificial intelligence. *Clinics in Dermatology*, *42*(3), 221-229. A "pre-history" overview, focusing on early ideas and milestones.
- Cao, Y., Liu, L., Li, M., Huang, Y., & Li, S. (2023). A comprehensive survey of AI-generated content (AIGC): A history of generative AI from GAN to ChatGPT. *arXiv preprint*. A somewhat technical but thorough overview of generative AI methods.
- Halevy, A., Norvig, P., & Pereira, F. (2009). The unreasonable effectiveness of data. *IEEE Intelligent Systems*, *24*(2), 8-12. A classic paper arguing that large datasets are often more important than sophisticated algorithms.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., … & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. In case you want to read the original paper. Not for the faint of heart.

Module 1 — How AI systems are designed

September 3, 2026:

- Lecture 03: Dataset design, labels, and tasks.
- Kahoot Quiz
- **Assignment 01**.

Suggested readings:

- Athalye, A., Northcutt, C., & Mueller, J. (2024). Dataset creation and curation in *Data-Centric AI*. A chapter from a forthcoming book on data-centric AI, covering the main ideas in dataset design.
- Rich, A. S., & Gureckis, T. M. (2019). Lessons for artificial intelligence from the study of natural stupidity. *Nature Machine Intelligence*, 1(4), 174-180. A title this good is hard to find. The article discusses different sources of bias in AI systems.
- Google PAIR. (2021). People + AI Guidebook: data collection and evaluation. How to plan a dataset, write labeller instructions, test inter-labeller reliability, and treat data workers fairly.

Further readings:

- Sambasivan, N., Kapania, S., Highfill, H., Akrong, D., Paritosh, P., & Aroyo, L. (2021). "Everyone wants to do the model work, not the data work": data cascades in high-stakes AI. *CHI 2021*. Interviews with 53 practitioners showing how small labelling problems compound downstream.
- Liang, W., Tadesse, G. A., Ho, D., Fei-Fei, L., Zaharia, M., Zhang, C., & Zou, J. (2022). Advances, challenges and opportunities in creating data for trustworthy AI. *Nature Machine Intelligence*, 4(8), 669-677. A good discussion on the challenges of creating high-quality datasets.
- Chai, C., & Li, G. (2020). Human-in-the-loop techniques in machine learning. *IEEE Data Eng. Bull.*, 43(3), 37-52. A little more technical, but quite comprehensive.

September 8, 2026:

- Lecture 04: Supervised, unsupervised, and reinforcement learning.
- Kahoot Quiz

Suggested readings:

- Jiang, T., Gradus, J. L., & Rosellini, A. J. (2020). Supervised machine learning: a brief primer. *Behavior Therapy*, *51*(5), 675-687. A non-technical introduction to supervised machine learning.
- Amazon Web Services (2025). What is the difference between supervised and unsupervised learning? A practical overview with examples.
- Ghasemi, M., & Ebrahimi, D. (2024). Introduction to reinforcement learning. *arXiv preprint*. Short introduction to the topic. If you would like to read something more advanced, please refer to Sutton & Barto (2015).

Further readings:

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Chapter 5: Machine learning basics in *Deep Learning*. A technical but clear introduction to the main ideas behind machine learning, by some of the founders of deep learning.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An introduction to statistical learning. A widely used book covering supervised and unsupervised learning methods. Included here for reference, as you'll surely use it in your future career.

September 10, 2026:

- Lecture 05: Metrics, validation and overfitting.
- Kahoot Quiz
- **Assignment 01 due (5%).**
- **Assignment 02**.

Suggested readings:

- Coursera (2025). Precision vs. recall in machine learning: What's the difference?. A short article about those two commonly confused metrics.
- Akinkugbe, A. (2025). The essential guide to model evaluation metrics for classification. A practical guide to common evaluation metrics.
- Thomas, R. L., & Uminsky, D. (2022). Reliance on metrics is a fundamental challenge for AI. *Patterns*, 3(5).100440. A discussion of the limitations of metrics.

Further readings:

- Confident AI (2025). LLM evaluation metrics: the ultimate LLM evaluation guide. A simple guide on new ways to measure the quality of LLM outputs.
- Rashka, S. (2025) Understanding the 4 main approaches to LLM evaluation. Good blog post about the different ways to evaluate LLMs.
- Inie, N., Stray, J., & Derczynski, L. (2023). Summon a demon and bind it: a grounded theory of LLM red teaming in the wild. *arXiv preprint*. A good overview of red teaming.
- Radharapu, B., Robinson, K., Aroyo, L., & Lahoti, P. (2023). Aart: AI-assisted red-teaming with diverse data generation for new LLM-powered applications. *arXiv preprint*. A method to improve LLM safety with simple adversarial attacks.
- Yee, E. (2025). AI red-teaming design: threat models and tools. *Center for Security and Emerging Technology*. A survey of 23 red-teaming tools showing how the threat model built into each one shapes the results you get.

Module 2 — Language and perception

September 15, 2026:

- Lecture 06: How machines handle language: tokens, embeddings and context.
- Kahoot Quiz

Suggested readings:

- StackOverflow (2023). An intuitive introduction to text embeddings. A practical introduction to embeddings.
- Metzger, S. (2022). A beginner's guide to tokens, vectors, and embeddings in NLP. Easy to read and with interesting examples.
- HuggingFace (2025). Tokenizer summary A really good overview of tokenisation, with examples and videos.

Further readings:

- Neptune.ai (2025). The ultimate guide to word embeddings. A clear and practical introduction to word embeddings.
- Neptune.ai (2025). Tokenization in NLP: types, challenges, examples, tools. Another article by the same group on tokenisation.
- Boykis, V. (2022). What are embeddings?. A whole book dedicated to embeddings. For those who are interested in a more in-depth treatment. There's also a GitHub repository with code examples.

September 17, 2026:

- Lecture 07: How machines see and hear.
- Kahoot Quiz
- **Assignment 02 due (5%).**
- **Assignment 03**.

Suggested readings:

- Gupta, A. (2025). How computers see the world: a beginner's guide to CNNs. An illustrated introduction to convolutional neural networks. Feel free to skip the coding section if you are not interested.
- AltexSoft (2025). How AI sound, music, and voice generation works. Another practical introduction. It goes a little beyond what we cover in class.

Further readings:

- O'Shea, K., & Nash, R. (2015). An introduction to convolutional neural networks. *arXiv preprint*. A clear and concise tutorial.
- Raschka, S. (2024). Understanding Multimodal LLMs. More technical but clear.
- Wu, J., Gan, W., Chen, Z., Wan, S., & Yu, P. S. (2023). Multimodal Large Language Models: A Survey. *arXiv preprint*. A broad survey of how models combine text with images, audio and video.
- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023, July). Robust speech recognition via large-scale weak supervision. In *International conference on machine learning* (pp. 28492-28518). PMLR. The original paper for Whisper.

September 22, 2026:

- Lecture 08: Review session for Quiz 01. This session covers Lectures 01-07 and prepares students for Quiz 01 on September 24.
- Kahoot Quiz

September 24, 2026:

- Lecture 09: **Quiz 01**. This quiz covers Lectures 01-07.
- **Assignment 03 due (5%).**
- **Assignment 04**.

September 29, 2026:

- Lecture 10: Prompting techniques.

Suggested readings:

- Anthropic (2025). Prompt engineering overview. Official guide from Claude's creators.
- Google (2025). Gemini for Workspace Prompting Guide. The persona-task-context-format (PTCF) framework.
- Prompting Guide. Community-maintained reference with examples.

Further readings:

- OpenAI (2025). Prompt engineering. OpenAI's best practices.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., … & Amodei, D. (2020). Language models are few-shot learners. *Advances in neural information processing systems*, 33, 1877-1901.
- Wei, J., et al. (2022). Chain-of-Thought prompting elicits reasoning in large language models. The original CoT paper.
- Google (2025). Prompting essentials. Free course on Coursera.
- Berryman, J., & Ziegler, A. (2024). Prompt engineering for LLMs: the art and science of building large language model–based applications. A recent book about the topic.

Module 3 — Retrieval, generation, pipelines and agents

October 1, 2026:

- Lecture 11: Creativity and hallucination.
- Kahoot Quiz
- **Assignment 04 due (5%).**
- **Assignment 05**.

Suggested readings:

- Jiang, X., Tian, Y., Hua, F., Xu, C., Wang, Y., & Guo, J. (2024). A survey on large language model hallucination via a creativity perspective. *arXiv preprint*. Very interesting paper. If you want to read just one article on this topic, this is a good candidate.

- Shao, A. (2025). New sources of inaccuracy? A conceptual framework for studying AI hallucinations. *Harvard Kennedy School Misinformation Review*. Worth reading as well.
- OpenAI (2024). Does ChatGPT tell the truth? Practical advice.

Further readings:

- Wang, C., Liu, X., Yue, Y., Guo, Q., Hu, X., Tang, X., … & Zhang, Y. (2025). Survey on factuality in large language models. *ACM Computing Surveys, 58(1), 1-37*. Wide-ranging and informative. Not technical.
- Anh-Hoang, D., Tran, V., & Nguyen, L. M. (2025). Survey and analysis of hallucinations in large language models: attribution to prompting strategies or model behavior. *Frontiers in Artificial Intelligence, 8, 1622292*. The authors argue that chain-of-thought prompting reduces hallucinations in prompt-sensitive scenarios.
- Luo, Q., King, G., Puett, M., & Smith, M. D. (2026). Inducing sustained creativity and diversity in large language models. *Working paper, Harvard University*. The authors propose a decoding method that produces more varied outputs than standard mode-seeking decoding.

October 6, 2026:

- Lecture 12: Retrieval-augmented workflows and semantic search.
- Kahoot Quiz

Suggested readings:

- Amazon Web Services (2025). What is retrieval-augmented generation (RAG)? Beginners' guide to RAG.
- Slack (2025). Semantic search explained: how it works, and why it matters. Good explainer.
- RAG Playground (2026). Interactive RAG pipeline visualisation. Hands-on tool for text splitting, embeddings and semantic search. We use it in class.

Further readings:

- Mastering LLM (2024). 11 chunking strategies for RAG — simplified & visualized. Nice explanation of chunking strategies.
- Willison, S. (2023). Embeddings: what they are and why they matter. A clear, beginner-friendly tour of embeddings and semantic search.
- Anthropic (2024). Introducing contextual retrieval. Technical, but shows how companies reduce retrieval failures.

October 8, 2026:

- Lecture 13: **Quiz 02**. This quiz covers Lectures 10-12.
- **Assignment 05 due (5%).**

October 13, 2026: Fall Break (No Classes)

October 15, 2026:

- Lecture 14: When AI systems fail: pipelines, monitoring, and documentation.
- **Assignment 06**.

Suggested readings:

- Spot Intelligence (2024). Data drift in machine learning. An accessible overview of how data drift affects ML systems over time.
- Howard, J. (2024). Context degradation syndrome: when large language models lose the plot. A discussion on why model performance degrades over time.
- Crawford, K., & Paglen, T. (2019). Excavating AI: The politics of images in machine learning training sets. A compelling visual essay on ImageNet's problems and the politics of training data.

Further readings:

- Google (2025). Production ML systems: monitoring pipelines. Practical guide to monitoring ML pipelines.
- Nawani, M. (2025). From correctness to confidence: understanding LLM testing. Explains the challenges of testing non-deterministic AI systems.

- ApX Machine Learning (2025). Input validation and sanitization for LLMs. Practical guide to defending AI systems against prompt injection attacks.
- Gebru, T., et al. (2018). Datasheets for datasets. *arXiv preprint*. The original paper proposing datasheets for datasets.
- Mitchell, M., et al. (2019). Model cards for model reporting. *Proceedings of FAccT*. The original paper proposing model cards.
- AI Incident Database. AI Incident Database. A collection of real-world AI failures and incidents for case study research.

October 20, 2026:

- Lecture 15: AI agents: when models start doing things.
- Kahoot Quiz

Suggested readings:

- IBM (2025). What are AI agents? A friendly explainer to start with.
- Heikkilä, M. (2024). What are AI agents? *MIT Technology Review*. A clear, non-technical overview of the agent idea and the hype around it.
- Willison, S. (2025). The lethal trifecta for AI agents. Why agents with private data, untrusted content, and the ability to communicate are a security nightmare. Short and very readable.
- Anthropic (2025). Project Vend: Can Claude run a small shop? Anthropic let an AI agent run their office shop for a month. It did not go well, and the write-up is honest and funny. Read this one.
- Kurzgesagt – In a Nutshell (2026). AI just crossed the terrifying line: now what? *YouTube*. An animated introduction to AI agents, starting with the agents that hacked Hugging Face.

Further readings:

- Anthropic (2024). Building effective agents. How Claude's creators think about agent design. Slightly technical, but the diagrams alone are worth your time.
- OpenAI (2025). A practical guide to building agents. OpenAI's guide for companies deploying agents. Skim the sections on guardrails.

October 22, 2026:

- Lecture 16: Setting up AI: instructions, memory and connectors.
- Kahoot Quiz
- **Assignment 06 due (5%).**
- **Assignment 07**.

Suggested readings:

- Anthropic (2026). How can I create and manage projects? *Claude Help Center*. The official walk-through of project instructions, knowledge files and the five-project limit on the free plan. The companion page on memory and chat search shows you how to read, edit and delete what Claude has saved about you.
- Chandrasekar, A. (2026). Your chatbot's memory of you can shape the information you see. *Columbia Journalism Review*, Tow Center. A short piece on what memory does to the answers you get, including memory poisoning and deleted memories that come back. Read it before you open your own memory panel.
- Anthropic (2024). Introducing the Model Context Protocol. The original announcement, written for a general audience. Useful for seeing what problem MCP was meant to solve before it became an industry standard.
- Agentic AI Foundation (2026). AGENTS.md. The whole specification is one page of plain Markdown with examples, which is the point. It is now used in tens of thousands of open-source projects and read by more than twenty different coding tools.

Further readings:

- Dash, A., Das, S., Kirsten, E., Wu, Q., Karnam, S. K., Gummadi, K. P., Holz, T., Zafar, M. B., & Zannettou, S. (2026). The algorithmic self-portrait: deconstructing memory in ChatGPT. *Proceedings of the ACM Web Conference 2026*. Eighty people donated their ChatGPT memories, 2,050 entries in all. The system wrote 96% of them without being asked, 28% held personal data under the GDPR definition, and 52% held psychological inferences.

- Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: compromising real-world LLM-integrated applications with indirect prompt injection. *arXiv preprint*. The paper that named indirect prompt injection and showed it working against Bing Chat. Once a tool fetches outside text, data and instructions look the same to the model.
- Hou, X., Zhao, Y., Wang, S., & Wang, H. (2025). Model Context Protocol (MCP): landscape, security threats, and future research directions. *arXiv preprint*. A survey of how MCP servers are built, deployed and maintained, with sixteen named attack scenarios. Technical in places, but the threat table reads on its own.
- Anthropic (2025). Effective context engineering for AI agents. An engineering post arguing that context is a finite resource, so the aim is the smallest set of high-signal instructions rather than the longest. It explains why a two-page instruction file often performs worse than ten good lines.

Module 4 — Data ethics and bias

October 27, 2026:

- Lecture 17: Types of bias and how they arise.
- Kahoot Quiz
- No assignment this week.
- Instructions for the final project (PDF). Groups must be finalised by November 3. Optional proposal due November 12. Final report and infographic due December 8.

Suggested readings:

- Silberg, J., & Manyika, J. (2019). Notes from the AI frontier: tackling bias in AI (and in humans). McKinsey Global Institute.
- Liu Roc, A. (2023). Understanding the impossibility of fairness. *Medium*. A good, accessible overview of the impossibility results in fairness.
- Kantayya, S. (2020). Coded Bias. Documentary film. May or may not be available on Netflix depending on your location.

Further readings:

- Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and machine learning. Chapter 2: When is automated decision making legitimate?
- Mehrabi, N., Morstatter, F., Saxena, N., Lettich, A., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, *54*(6), 1-35.
- Hellman, D. (2025). Algorithmic fairness. *The Stanford Encyclopedia of Philosophy* (Fall 2025 Edition).

Module 5 — Policy, governance and social impact

October 29, 2026:

- Lecture 18: AI regulation and standards around the world.
- Kahoot Quiz
- **Assignment 07 due (5%).**

Suggested readings:

- European Parliament (2023). EU AI act. The original text.
- Wikipedia (2026). Regulation of artificial intelligence. It's hard to be updated on this topic, so this Wikipedia page is a good resource to keep track of the latest developments.
- The White House (2025). Winning the race: America's AI action plan. What the US government wants to do about AI.

Further readings:

- GOV.UK (2023). A pro-innovation approach to AI regulation.. The UK's approach to AI regulation.
- Cole, M. D. (2024). AI regulation and governance on a global scale. *Journal of AI Law and Regulation*, *1*(1), 126-142. A short report on national and international efforts to regulate AI.
- G'Sell, F. (2025). Regulating under uncertainty: governance options for generative AI. A report on the challenges of regulating AI and possible approaches to do it effectively.

November 3, 2026:

- Lecture 19: Privacy and data protection.
- Kahoot Quiz
- **Group list due.**

Suggested readings:

- Taddeo, M., & Floridi, L. (2018). How AI can be a force for good. *Science*, *361*(6404), 751-752. A philosophical discussion on the ethical implications of AI, including privacy concerns.
- OpenAI (2025). OpenAI privacy portal. OpenAI's privacy policy and practices.
- King, J. & Meinhardt, C. (2024). Rethinking privacy in the AI era: policy provocations for a data-centric world. The authors analyse how existing and future privacy and data protection regulation will impact the development and deployment of AI systems.

Further readings:

- Hoofnagle, C. J., Van Der Sloot, B., & Borgesius, F. Z. (2019). The European Union general data protection regulation: what it is and what it means. *Information & Communications Technology Law*, *28*(1), 65-98. A detailed overview of the GDPR, the most influential data protection law in the world.
- European Data Protection Board (2025). AI privacy risks & mitigations Large Language Models (LLMs). This report provides use cases examples on the application of the risk management framework in real-world scenarios.
- OECD (2026). OECD due diligence guidance for responsible AI. Practical guidance for organisations to implement good AI practices.
- Future of Privacy Forum (2025). `https://fpf.org/`. An NGO that advances principled and pragmatic data protection, AI and digital governance practices. Many resources on their website.

November 5, 2026:

- Lecture 20: **Quiz 03**. This quiz covers Lectures 14-18.
- **Assignment 08**.

November 10, 2026:

- Lecture 21: Labour markets and economic effects.

Suggested readings:

- Financial Times. (2026). The great graduate job drought. Also available here. An overview of the job market for recent graduates. You may want to read this one. Spoiler: it's not good.
- Oks, D. (2026). Why the ATM didn't kill bank teller jobs (but the iPhone did). A short essay arguing that task-automating technologies rarely eliminate jobs, while paradigm-shifting ones can. The author applies this distinction to AI.
- Karpathy, A. (2025). AI exposure across US occupations. An interactive visualisation of AI exposure scores for 342 US occupations, with breakdowns by education level, pay band, and employment outlook.

Further readings:

- Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: how technology displaces and reinstates labor. *Journal of Economic Perspectives*, *33*(2), 3-30.
- World Economic Forum. (2025). The future of jobs report 2025. A wide-ranging report on how jobs are expected to change.
- Jones, C. (2026). AI and our economic future. A working paper that reviews the literature on the economic effects of AI. Written for a broad audience, to be published in the *Journal of Economic Perspectives*.
- Strain, Michael R. (2024). The case for AI optimism.. Maybe things as not as bad as they seem?
- Anthropic (2026). Anthropic economic index. How people use Claude and what are its possible impacts on the economy.
- OpenAI (2025). How people are using ChatGPT. Same thing, but for ChatGPT. Spoiler: most people are not using it for work. The full article is available here.

Module 6 — Applications, limits and projects

November 12, 2026:

- Lecture 22: AI and wellbeing: the attention economy, mental health, and the environment.
- Kahoot Quiz
- **Assignment 08 due (5%).**
- **Assignment 09**.
- **Optional project proposal due.**

Suggested readings:

- OpenAI. (2025). Strengthening ChatGPT's responses in sensitive conversations. OpenAI's approach to making ChatGPT more helpful in sensitive conversations.
- McBain, R. K., et al. (2025). Use of generative AI for mental health advice among adolescents and young adults. *JAMA Network Open*. A nationally representative survey showing 1 in 8 US adolescents and young adults use AI chatbots for mental health advice.
- MIT Technology Review (2025). We did the math on AI's energy footprint. Here's the story you haven't heard. A data-driven investigation into the opacity of Big Tech's energy reporting.

Further readings:

- Habicht, J., Dina, L. M., McFadyen, J., Stylianou, M., Harper, R., Hauser, T. U., & Rollwage, M. (2025). Generative AI–enabled therapy support tool for improved clinical outcomes and patient engagement in group therapy: real-world observational study. *Journal of Medical Internet Research*, *27*, e60435. An experiment on the use of AI in group therapy. It has a positive result.
- Moore, J., Grabb, D., Agnew, W., Klyman, K., Chancellor, S., Ong, D. C., & Haber, N. (2025). Expressing stigma and inappropriate responses prevents LLMs from safely replacing mental health providers. *arXiv preprint arXiv:2504.18412*. The authors question the idea that LLMs should replace mental health providers.
- Habicht, J., et al. (2024). Closing the accessibility gap to mental health treatment with a personalised self-referral chatbot. *Nature Medicine*. A study of 129,400 NHS patients showing a chatbot self-referral system increased mental health referrals by 15% overall and 29% among ethnic minorities.

- Orben, A., & Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. *Nature Human Behaviour*, *3*, 173-182. A specification-curve analysis of 355,000+ adolescents finding that technology use explains at most 0.4% of variation in wellbeing.
- Twenge, J. M., Haidt, J., Lozano, J., & Cummins, K. M. (2022). Specification curve analysis shows that social media use is linked to poor mental health, especially among girls. *Acta Psychologica*, *224*, 103512. A direct rebuttal to Orben & Przybylski using the same method, arguing the relationship is stronger and gender-differentiated.
- Arguedas, A. R., Robertson, C. T., Fletcher, R., & Nielsen, R. K. (2022). Echo chambers, filter bubbles, and polarisation: a literature review. Reuters Institute, University of Oxford. A careful synthesis of the evidence, finding the picture is more mixed than popular coverage suggests.

November 17, 2026:

- Lecture 23: Misinformation, deepfakes and trust online.
- Kahoot Quiz

Suggested readings:

- Chesney, R., & Citron, D. K. (2019). Deepfakes and the new disinformation war. *Foreign Affairs*. A short article about the risks of deepfakes and how to address them.
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, *359*(6380), 1146-1151. A seminal paper on the topic of misinformation. It shows that false news spreads faster and farther than true news, and that bots are not the main driver of this phenomenon.
- Helmus, T. C. (2022). Artificial intelligence, deepfakes, and disinformation: a primer. *RAND Corporation*. A major revision of the literature.

Further readings:

- Lazer, D. M., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., … & Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380), 1094-1096. A call to action for researchers to study misinformation and develop solutions.

- Allcott, H., & Gentzkow, M. (2017). Social media and fake news in the 2016 election. *Journal of economic perspectives*, 31(2), 211-236. A widely-cited paper that analyses the economics of fake news.
- Mirsky, Y., & Lee, W. (2020). The creation and detection of deepfakes: a survey. *arXiv preprint* arXiv:2004.11138. A technical survey on deepfakes. Quite technical, though. If you are interested in the topic but don't have a technical background, you can skip this one and read the next one instead.

November 19, 2026:

- Lecture 24: **Quiz 04**. This quiz covers Lectures 19, 21, and 22 (Privacy, Labour, and Wellbeing).
- **Assignment 09 due (5%).**
- **Assignment 10**.

November 24, 2026:

- Lecture 25: Long-term safety, the alignment problem and the future of AI.

Suggested readings:

- Russell, S. (2017). 3 principles for creating safer AI. *TED Talk*.
- Anthropic. (2025). Constitutional classifiers: defending against universal jailbreaks. A jailbreak defense system using input/output classifiers trained on synthetic data. It reduced jailbreak success rates by a lot… but some people still found ways around it. If you would like to read the whole academic paper, check it out here.
- Center for AI Safety. A non-profit organisation working on this topic. They have a lot of resources on their website.
- The New York Times. (2026). How terrorist groups are using A.I. to gain an edge in battle. An investigation into how Boko Haram fighters in Nigeria used AI chatbots to plan attacks, based on interviews with former members. Also available here.

Further readings:

- Amodei, D., et al. (2016). Concrete problems in AI safety. *arXiv preprint*. A highly influential paper on the practical challenges of AI safety.
- Ji, J., Qiu, T., Chen, B., Zhang, B., Lou, H., Wang, K., … & Gao, W. (2023). AI alignment: a comprehensive survey. *arXiv preprint*. A survey of the field of AI alignment.
- Zhi-Xuan, T., Carroll, M., Franklin, M., & Ashton, H. (2025). Beyond preferences in AI alignment. *Philosophical Studies*, 182(7), 1813-1863. A philosophical take on the limits of rational choice in AI alignment. A little more abstract, but very interesting.
- Christian, B. (2024). The alignment problem: machine learning and human values. *Penguin Random House*. A great book on the challenges of aligning AI systems with human values.
- AI Alignment Forum. A community website for discussing AI alignment. Many of the most important papers in the field are posted here.

November 26, 2026: Thanksgiving Recess (No Classes)

December 1, 2026:

- Lecture 26: Course revision.
- Kahoot Quiz

December 3, 2026:

- Lecture 27: **Quiz 05**. This quiz covers Lectures 23 and 25 (Misinformation and Safety).
- **Assignment 10 due (5%).**

December 8, 2026:

- No lecture.
- Q&A session and final project work time.
- Final project due (20%).